

OAI 0.2.3 08/01/17 Subject characteristics, risk factors Release Comments

Variable	Label	Comment	SAS Code
P01BROHRCV	Brother had hip replacement surgery (calc)		Collaps2mcr.sas
			MOMHRtoBROHRCVG.sas
P01BROKRCV	Brother had knee replacement surgery (calc)		Collaps2mcr.sas
			MOMKRtoBROKRCVG.sas
P01DADHRCV	Father had hip replacement surgery (calc)		Collaps2mcr.sas
			MOMHRtoBROHRCVG.sas
P01DADKRCV	Father had knee replacement surgery (calc)		Collaps2mcr.sas
			MOMKRtoBROKRCVG.sas
P01LXRKOA	Left knee baseline radiographic OA (definite osteophytes, calc, used in OAI definition of symptomatic knee OA)	Data change alert for 0.2.1: As of release 0.2.1, LXRKOA, LXRKOA2, RXRKO, RXRKO2 and XRKOA are assigned to the Biomarkers00 dataset. They were previously assigned to the JointSx00 dataset.	RKSX_LKSXG.sas
			SxOA.sas
P01LXRKOA2	Left knee baseline x-ray: osteophytes and JSN (calc)	See comment for variable P01LXRKOA ("Data change alert for 0...")	RKSX_LKSXG.sas
			SxOA.sas
P01MOMHRCV	Mother had hip replacement surgery (calc)		Collaps2mcr.sas
			MOMHRtoBROHRCVG.sas
P01MOMKRCV	Mother had knee replacement surgery (calc)		Collaps2mcr.sas
			MOMKRtoBROKRCVG.sas

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Variable	Label	Comment	SAS Code
P01MRKCBCV	Knees fit comfortably inside knee coil with coil completely closed (calc)	Data change alert for 0.2.1: As of release 0.2.1, the format for the variable, V00MRKCBCV, has been corrected. The format has been corrected so that the value labels for 4 and 5 are as follows: 4: Yes, left knee only right knee TKR and 5: Yes, right knee only, left knee TKR. In release 0.1.1, the format was incorrect.	MRKCBCVG.sas
P01MRPRBCV	Any problems related to the MRI scan (calc)		Collaps2mcr.sas
			MRPRB.sas
P01OAGRDL	Left knee baseline x-ray: composite OA grade (calc)		CompositeOA.sas
P01OAGRDR	Right knee baseline x-ray: composite OA grade (calc)		CompositeOA.sas
P01RXRKO	Right knee baseline radiographic OA (definite osteophytes, calc, used in OAI definition of symptomatic knee OA)	See comment for variable P01LXRKO ("Data change alert for 0...")	RKSX_LKSXG.sas
			SxOA.sas
P01RXRKO2	Right knee baseline x-ray: osteophytes and JSN (calc)	See comment for variable P01LXRKO ("Data change alert for 0...")	RKSX_LKSXG.sas
			SxOA.sas
P01SISHRCV	Sister had hip replacement surgery (calc)		Collaps2mcr.sas
			MOMHRtoBROHRCVG.sas
P01SISKRCV	Sister had knee replacement surgery (calc)		Collaps2mcr.sas
			MOMKRtoBROKRCVG.sas
P01SVLKJSL	Left knee baseline x-ray: evidence of knee lateral joint space narrowing (calc)	If 2 x-rays were taken, the values from the released image are used.	

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Variable	Label	Comment	SAS Code
P01SVLKJSM	Left knee baseline x-ray: evidence of knee medial joint space narrowing (calc)	See comment for variable P01SVLKJSL ("If 2 x rays were...")	
P01SVLKMI	Left knee baseline x-ray: evidence of knee metal artifacts (calc)	See comment for variable P01SVLKJSL ("If 2 x rays were...")	
P01SVLKOST	Left knee baseline x-ray: evidence of knee osteophytes (calc)	See comment for variable P01SVLKJSL ("If 2 x rays were...")	
P01SVRKJSL	Right knee baseline x-ray: evidence of knee lateral joint space narrowing (calc)	See comment for variable P01SVLKJSL ("If 2 x rays were...")	
P01SVRKJSM	Right knee baseline x-ray: evidence of knee medial joint space narrowing (calc)	See comment for variable P01SVLKJSL ("If 2 x rays were...")	
P01SVRKMI	Right knee baseline x-ray: evidence of knee metal artifacts (calc)	See comment for variable P01SVLKJSL ("If 2 x rays were...")	
P01SVRKOST	Right knee baseline x-ray: evidence of knee osteophytes (calc)	See comment for variable P01SVLKJSL ("If 2 x rays were...")	
P01SVXRELK	Baseline knee x-ray: knee(s) eligible for MRI based on x-ray (calc)	See comment for variable P01SVLKJSL ("If 2 x rays were...")	
P01SVXRLKR	Left knee baseline x-ray: evidence of knee replacement (calc)	See comment for variable P01SVLKJSL ("If 2 x rays were...")	
P01SVXRRID	Baseline knee x-ray: clinical center reader ID # (calc)	See comment for variable P01SVLKJSL ("If 2 x rays were...")	
P01SVXRRKR	Right knee baseline x-ray: evidence of knee replacement (calc)	See comment for variable P01SVLKJSL ("If 2 x rays were...")	

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Variable	Label	Comment	SAS Code
P01XRKOA	Baseline radiographic knee OA status by person (calc)	See comment for variable P01LXRKOA ("Data change alert for 0...")	RKSX_LKSXG.sas
			SxOA.sas
P02ACTRISK	Engage in at least one frequent knee bending activity (calc, used for study eligibility initially)	Engaging in any one of four frequent knee bending activities was used as a qualifying risk factor initially in OAI. When it became clear that the prevalence of this risk factor was considerably higher than the limit set for the prevalence of any one risk factor, frequent knee bending was dropped as a qualifying risk factor. However, a number of participants who would not otherwise have been eligible had already been enrolled by virtue of reporting a frequent knee bending activity.	ACTRISK.sas
P02ELGRISK	Knee symptoms, risk factors, or both, status at IEI/SV (calc)		ELGRISK.sas
P02IKPRISK	Either knee symptom status at IEI (calc, used for study eligibility)		IKPRISK.sas
P02JBMPCV	Hard bumps on fingers of right hand, left hand or both hands (calc, if=3 used for study eligibility)		JBMPCV.sas
P02KRS3CV	Which knee considering having replaced, next 3 years (calc)		KRS3CV.sas
P02STMEDCV	Told what study medications taken for research study (calc)		STMED.sas
P02WTGA	Above weight cut-off for age/gender group (calc, used for study eligibility)		WTGA.sas
V00AGE	Age (calc, used for study eligibility)		AGEG.sas
V00CEMPLOY	Current employment (calc)		EmployG.sas
V00EDCV	Highest grade or year of school completed (calc)		LTHS.sas

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Variable	Label	Comment	SAS Code
		The refused response option (.R: refused) is not included as an option for unreleased variables during the administration of the Self-Administered Questionnaire form. This response can be coded by the clinical centers to account for missing data for the unreleased variable used in the calculation of this collapsed variable. The refused response option for collapsed variables has been added to the annotated data collection forms.	
V00ENREXCP	Participant has enrollment exception (calc)	Data change alert for 0.2.1: As of release 0.2.1, the number of enrollment exception forms has changed. One participant in release 0.1.1 now has an enrollment exception form entered and several participants in release 0.1.1 had their enrollment exception form deleted from the data system. Enrollment exception forms were deleted when the form was entered in error (the participant did not require an enrollment exception form because they had met all the enrollment criteria). These changes were a result of the process of reconciling the final list of OAI enrolled participants.	EnrExcpt.sas
V00HANDED	Dominant hand for hand x-ray (calc)		handed.sas
V00HLTHCAR	Where usually go for health care or advice about health care	The refused response option (.R: refused) is not included as an option on the Self-Administered Questionnaire form. However, this response can be coded in the data system for this variable by the clinical centers to account for missing data.	
V00HLTHCOV	Currently have any kind of health care coverage	See comment for variable V00HLTHCAR ("The refused response option R:...")	
V00HOUACT1	Household activities: light housework, past 7 days	Data change alert for 0.2.1 and 1.1.2: As of release 0.2.1 and 1.1.2, some PASE scores have changed as a result of a change in how the score is calculated. In the revised code, if hours per day items and/or the occupational activity items are missing, then the median of the item is obtained using all available data for that visit. This median is now used in place of the missing data to calculate the PASE score. If the obtained median is a decimal number, it is rounded down. If there are more than 3 items missing for the hours per day/occupational activity then the PASE score is set to missing. If any of the activity frequency items, household items or key work items are missing, the PASE score is set to missing. Also, WORKHR7 and WORK1HR are now unreleased and replaced by WKHR7CV. The Physical Activity Scale for the Elderly (PASE®) is a copyrighted instrument that may not be displayed. Please go to: https://www.neriscience.com/Resources/Products/PASE for more information about the PASE® Information about PASE® scoring for baseline can be found at the link below: http://www.oai.ucsf.edu/datarelease/AdditionalVarComments/PASE.pdf Information about PASE® scoring for follow-up can be found at the link below: http://www.oai.ucsf.edu/datarelease/AdditionalVarComments/PASEFU.pdf	PASEG.sas
V00HOUACT2	Household activities: heavy housework, past 7 days	See comment for variable V00HOUACT1 ("Data change alert for 0...")	PASEG.sas

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Variable	Label	Comment	SAS Code
V00HOUACT3	Household activities: home repairs, past 7 days	See comment for variable V00HOUACT1 ("Data change alert for 0...")	PASEG.sas
V00HOUACT4	Household activities: lawn work/yard care, past 7 days	See comment for variable V00HOUACT1 ("Data change alert for 0...")	PASEG.sas
V00HOUACT5	Household activities: outdoor gardening, past 7 days	See comment for variable V00HOUACT1 ("Data change alert for 0...")	PASEG.sas
V00HOUACT6	Household activities: caring for another person, past 7 days	See comment for variable V00HOUACT1 ("Data change alert for 0...")	PASEG.sas
V00INCOME	Yearly income (calc)	The refused response option (.R: refused) is not included as an option for the unreleased income variables during the administration of the Self-Administered Questionnaire form. This response can be coded by the clinical centers to account for missing data for the unreleased variables used in the calculation of these calculated variables.	IncomeG.sas
V00INCOME2	Yearly income (>50K or <50K) (calc)	See comment for variable V00INCOME ("The refused response option R:...")	IncomeG.sas
V00LIVE1	Live with spouse	See comment for variable V00HLTHCAR ("The refused response option R:...")	
V00LIVE2	Live with romantic partner	See comment for variable V00HLTHCAR ("The refused response option R:...")	
V00LIVE3	Live with children	See comment for variable V00HLTHCAR ("The refused response option R:...")	
V00LIVE4	Live with other relatives or friends	Data change alert for 0.2.1: For release 0.2.1, LV45CV has been unreleased and is replaced by LIVE4 and LIVE5.	
		See comment for variable V00HLTHCAR ("The refused response option R:...")	
V00LIVE5	Live with someone else (not child, spouse, romantic partner)	See comment for variable V00LIVE4 ("Data change alert for 0...")	
		See comment for variable V00HLTHCAR ("The refused response option R:...")	
V00LIVENO	Besides self, how many other people live in household (calc)	Data change alert for 0.2.1: As of release 0.2.1, the values for LIVENO were updated resulting in a change of values for some participants. Value "7: More than 6" has been removed and value "6: six" has been revised to "6: Six or more." Thus, some people have changed from LIVENO=7 to LIVENO=6 as of release 0.2.1. Also, LIVEALO has been unreleased and is replaced by LIVENO.	LIVENOG.sas
V00MARITST	Marital status	Data change alert for 0.2.1: For release 0.2.1, MARSTCV has been unreleased and is replaced by MARITST.	
		See comment for variable V00HLTHCAR ("The refused response option R:...")	

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Variable	Label	Comment	SAS Code
V00MEDINS	Have any health insurance plan that pays for all or part of cost of prescription medicines	See comment for variable V00HLTHCAR ("The refused response option R:...")	
V00PA130CV	How often climb up total of 10 or more flights of stairs during typical week, past 30 days (calc)		Collaps2mcr.sas
			PA130CVto530CVG.sas
V00PA130NM	How many flights of stairs have you climbed up, past 7 days (one flight equals 10 steps)	This question (PA130NM) was asked at baseline and 12-month follow-up visit with the following response options: Less than one flight. 1 to 2 flights. 3 to 4 flights. 5 to 6 flights. More than 6 flights. Don't know. Refused. This question (PA130NF) was asked starting at the 24-month follow-up visit with the following revised response options: Less than one flight. 1 to 2 flights. 3 to 4 flights. 5 to 6 flights. 7 to 8 flights. 9 to 14 flights. 15 to 21 flights. > 21 flights. Don't know. Refused.	
V00PA230CV	How often kneel 30 minutes or more during typical week, past 30 days (calc)		Collaps2mcr.sas
			PA130CVto530CVG.sas
V00PA330CV	How often squat for 30 minutes or more during typical week, past 30 days (calc)		Collaps2mcr.sas

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Variable	Label	Comment	SAS Code
			PA130CVto530CVG.sas
V00PA430CV	How often lift or move objects weighing 25 pounds or more by hand during a typical week, past 30 days (calc)		Collaps2mcr.sas
			PA130CVto530CVG.sas
V00PA530CV	How often get in and out of squatting position 10 or more times during a typical week, past 30 days (calc)		Collaps2mcr.sas
			PA130CVto530CVG.sas
V00PASE	Physical Activity Scale for the Elderly (PASE) score (calc)	See comment for variable V00HOUACT1 ("Data change alert for 0...")	PASEG.sas
V00PASE1	Leisure activities: sitting, past 7 days	The Physical Activity Scale for the Elderly (PASE®) is a copyrighted instrument that may not be displayed. Please go to: https://www.neriscience.com/Resources/Products/PASE for more information about the PASE® Information about PASE® scoring for baseline can be found at the link below: http://www.oai.ucsf.edu/datarelease/AdditionalVarComments/PASE.pdf Information about PASE® scoring for follow-up can be found at the link below: http://www.oai.ucsf.edu/datarelease/AdditionalVarComments/PASEFU.pdf	
V00PASE1HR	Leisure activities: sitting, hours per day, past 7 days	See comment for variable V00PASE1 ("The Physical Activity Scale for...")	
V00PASE2	Leisure activities: walking, past 7 days	See comment for variable V00HOUACT1 ("Data change alert for 0...")	PASEG.sas
V00PASE2HR	Leisure activities: walking, hours per day, past 7 days	See comment for variable V00HOUACT1 ("Data change alert for 0...")	PASEG.sas
V00PASE3	Leisure activities: light sport/recreation, past 7 days	See comment for variable V00HOUACT1 ("Data change alert for 0...")	PASEG.sas

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Variable	Label	Comment	SAS Code
V00PASE3HR	Leisure activities: light sport/recreation, hours per day, past 7 days	See comment for variable V00HOUACT1 ("Data change alert for 0...")	PASEG.sas
V00PASE4	Leisure activities: moderate sport/recreation, past 7 days	See comment for variable V00HOUACT1 ("Data change alert for 0...")	PASEG.sas
V00PASE4HR	Leisure activities: moderate sport/recreation, hours per day, past 7 days	See comment for variable V00HOUACT1 ("Data change alert for 0...")	PASEG.sas
V00PASE5	Leisure activities: strenuous sport/recreation, past 7 days	See comment for variable V00HOUACT1 ("Data change alert for 0...")	PASEG.sas
V00PASE5HR	Leisure activities: strenuous sport/recreation, hours per day, past 7 days	See comment for variable V00HOUACT1 ("Data change alert for 0...")	PASEG.sas
V00PASE6	Leisure activities: muscle strength/endurance, past 7 days	See comment for variable V00HOUACT1 ("Data change alert for 0...")	PASEG.sas
V00PASE6HR	Leisure activities: muscle strength/endurance, hours per day, past 7 days	See comment for variable V00HOUACT1 ("Data change alert for 0...")	PASEG.sas
V00WKHR7CV	Occupational activities: number of hours worked, past 7 days (calc)	The Physical Activity Scale for the Elderly (PASE®) is a copyrighted instrument that may not be displayed. Please go to: http://www.neriscience.com/Products/ProductDetail/tabid/212/ArticleId/45/Physical-Activity-Scale-for-the-Elderly-PASE.aspx for more information about the PASE® Information about PASE® scoring for baseline can be found at the link below: http://www.oai.ucsf.edu/datarelease/AdditionalVarComments/PASE.pdf Information about PASE® scoring for follow-up can be found at the link below: http://www.oai.ucsf.edu/datarelease/AdditionalVarComments/PASEFU.pdf	WKHRCVG.sas

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Variable	Label	Comment	SAS Code
V00WORK7	Occupational activities: work (pay/volunteer), past 7 days	See comment for variable V00HOUACT1 ("Data change alert for 0...")	PASEG.sas
V00WORKAMT	Occupational activities: occupational activity level, past 7 days	See comment for variable V00HOUACT1 ("Data change alert for 0...")	PASEG.sas

```

*****;
* Study: OAI *;
* Source name: ACTRISK.sas *;
* Source type: Program *;
* Author: Emily Kenyon *;
* Created: 04/06/2007 *;
* Input: daf.IEI (Initial Eligibility Interview) *;
* Output: P02ACTRISK *;
* Summary: Determines whether participant participated in *;
*           any one of four knee bending activities on a *;
*           frequent basis (was considered risk factor for *;
*           eligibility at first, later discarded *;
* *;
* Revisions: None to date *;
*****;

**** 0 = No
      1 = Yes
      .D = Don't know
      .R = Refused
      .M = Missing
      .A = Not expected
****;

      if p02pa1=1 or p02pa2=1 or p02pa3=1 or p02pa4=1 then P02ACTRISK=1;
      else if p02pa1=0 and p02pa2=0 and p02pa3=0 and p02pa4 =0 then P02ACTRISK=0;
      else if p02pa1=.R or p02pa2=.R or p02pa3=.R or p02pa4=.R then P02ACTRISK=.R;
      else if p02pa1=.D or p02pa2=.D or p02pa3=.D or p02pa4=.D then P02ACTRISK=.D;
      else if p02pa1=.M or p02pa2=.M or p02pa3=.M or p02pa4=.M then P02ACTRISK=.M;
      else if p02pa1=.A or p02pa2=.A or p02pa3=.A or p02pa4=.A then P02ACTRISK=.A;

```

```

*****;
* Study: OAI
* Source name: AGE.G.sas
* Source type: Program
* Author: Maurice Dockrell
* Created: 09/25/2006
* Input: &_lib..&_Mmo (Clinic Visit)
*       DAF.IEI (Initial Eligibility Interview)
* Output: AGE dataset
* Summary: Calculate age at Clinic Visit using date of
*          Birth from Q6 p2 of the IEI (P02DOB) and date of Clinic Visit
*          (EVDATE from Enrollment Visit Checklist, FVDATE from Follow-Up Visit Checklist)

* Revisions: Updated to run for all years, including baseline MD 03/08/07

*****;
*This code is written for use with data from any dataset, so it uses generic macro
variables to take the place of library, dataset prefixes and variable prefixes that change f
rom year to year, as follows:
    &_lib. stands for the SAS library containing the data for the year being processed
    (e.g., daf, daf1, daf2, etc)
    &_Mmo. stands for the visit prefix for the dataset (e.g. EV, M12, M24, etc)
    &_v. stands for the variable prefixes V00, V01, V02, etc.;

data age(keep=ID &_v.AGE);
%macro agecalc;
    %if &_N=0 %then %do;
        merge &_lib..&_Mmo(in=inds keep=ID &_v.EVDATE)
    %end;
    %else %do;
        merge &_lib..&_Mmo(in=inds keep=ID &_v.FVDATE)
    %end;
    daf.iei(in=iniei keep=ID P02DOB);
    by id;
    if inds and iniei;

**strip time indicators from date variable before calculating;
%if &_N=0 %then %do;
    &_v.EVDATE=datepart(&_v.EVDATE);
%end;
%else %do;
    &_v.FVDATE=datepart(&_v.FVDATE);
%end;
    P02DOB=datepart(P02DOB);

**format dates;
%if &_N=0 %then %do;
format &_v.EVDATE p02dob mmddyy10.;
%end;
%else %do;
format &_v.FVDATE p02dob mmddyy10.;
%end;

**handle missing values;
%if &_N=0 %then %do;
if &_v.EVDATE<=.z then &_v.AGE=&_v.EVDATE;
%end;
%else %do;
if &_v.FVDATE<=.z then &_v.AGE=&_v.FVDATE;
%end;
    else if P02DOB<=.z then &_v.AGE=P02DOB;

**calculate age;
    else do;
%if &_N=0 %then %do;
    &_v.AGE=floor(yrdif(P02DOB, &_v.EVDATE, "ACT/ACT"));
%end;
%else %do;

```

```
&_v.AGE=floor(yrdif(P02DOB, &_v.FVDATE, "ACT/ACT"));
%end;
    end;
    *(ACT/ACT - Actual days each month, Actual days each year in SAS);

%mend agecalc;
%agecalc;
run;
```

```

*****;
* Study: OAI *;
* Source name: Collaps2mcr.sas *;
* Source type: MACRO *;
* Author: Laura Akin *;
* Created: 03/31/2005 *;
* Input: see below *;
* Output: recoded variable *;
* Summary: see below *;
* Revisions: *;
* 01/23/2006: Format assignment removed. LAkin *;
*****;

*****
*;
** Macro Collaps2 *
*;
** *
*;
** Collapsing Main and Subvars *
*;
** MainVar=the "Parent" variable *
*;
** SubVar=the "Child" variable *
*;
** CalcVar=the newly created "calculated" variable *
*;
** *
*;
*****
*;
%macro Collaps2(mainvar,subvar,calcvar);
  length &calcvar 8;

  if &mainvar=0 then &calcvar=0;
  else if &mainvar<=.z then &calcvar=&mainvar;
  else if &mainvar=1 then &calcvar=&subvar;

%mend Collaps2;

***** END OF PROGRAM *****
*;

```

```

*****;
* Study: OAI *;
* Source name: CompositeOA.sas *;
* Source type: Program *;
* Author: Jason Maeda *;
* Created: 06/22/2007 *;
* Input: &_lib.&_Mmo (Screening Visit workbook) *;
* Variables: &_V.SVLKOST, &_V.SVLKJSM, &_V.SVLKJSL, &_V.SVRKOST, &_V.SVRKJSM, *;
*           &_V.SVRKJSL, &_V.SVXRRKR, &_V.SVXRRKR *;
* Output: &_V.OAGRDR, &_V.OAGRDL *;
* Summary: Calculates a composite OA score which can be used as a simulated KL grade *;

*
* Grade 0: Normal: No osteophytes or JSN *;
* Grade 1: Doubtful: possible or minute osteophyte of doubtful significance, *;
*           or isolated mild-moderate JSN where osteophyte=0 or 1 (questionable) *;
*           and JSN=1 *;
* Grade 2: Minimal: definite osteophytes and unimpaired joint space, or isolated *;
*           severe JSN where osteophyte=0 or 1 (questionable) and JSN=2 *;
* Grade 3: Moderate: definite osteophytes, moderate diminution of joint space *;
*           (OARSI grade 1-2) *;
* Grade 4: Severe: definite osteophytes, joint space greatly impaired (OARSI grade 3) *;
*
* This code is written for use with data from any dataset, so *;
* it uses generic macro variables to take the place of library, dataset *;
* prefixes and suffixes, and variable prefixes that change from year *;
* to year, as follows: *;
* &_lib. stands for the SAS library containing the data for the year being *;
* processed (e.g., daf, daf1, daf2, etc) *;
* &_Mmo. stands for the visit prefix for the dataset (e.g., EV, M12, M24, etc) *;
* &_N. stands for the visit suffix for the dataset, if one is used *;
* (e.g., 0, 1, 2, etc) *;
* &_v. stands for the variable prefixes V01, V01, V02, etc. *;
* &_visit. stands for the visit designation used in the dataset *;
* (e.g., 0, 1, 2, 66, etc) *;
*
* Revisions: None to date *;
*****;

if &_V.SVRKOST=0 and &_V.SVRKJSM=0 and &_V.SVRKJSL=0 then &_V.OAGRDR = 0 ;
else if &_V.SVRKOST=1 and &_V.SVRKJSM=0 and &_V.SVRKJSL=0 then &_V.OAGRDR = 1 ;
else if &_V.SVRKOST=2 and &_V.SVRKJSM=0 and &_V.SVRKJSL=0 then &_V.OAGRDR = 2 ;
else if &_V.SVRKOST=2 and &_V.SVRKJSM in (0,1) and
    &_V.SVRKJSL in (0,1) and
    not (&_V.SVRKJSM=0 and &_V.SVRKJSL=0) then &_V.OAGRDR = 3 ;
else if &_V.SVRKOST=2 and (&_V.SVRKJSM=2 or &_V.SVRKJSL=2) then &_V.OAGRDR = 4 ;
else if &_V.SVRKOST=0 and &_V.SVRKJSM in (0,1) and
    &_V.SVRKJSL in (0,1) and
    not (&_V.SVRKJSM=0 and &_V.SVRKJSL=0) then &_V.OAGRDR = 1 ;
else if &_V.SVRKOST=1 and &_V.SVRKJSM in (0,1) and
    &_V.SVRKJSL in (0,1) and
    not (&_V.SVRKJSM=0 and &_V.SVRKJSL=0) then &_V.OAGRDR = 1 ;
else if &_V.SVRKOST=1 and (&_V.SVRKJSM=2 or &_V.SVRKJSL=2) then &_V.OAGRDR = 2 ;
else if &_V.SVRKOST=0 and (&_V.SVRKJSM=2 or &_V.SVRKJSL=2) then &_V.OAGRDR = 2 ;
else if &_V.SVRKOST in (.,.M) or &_V.SVRKJSM in (.,.M) or
    &_V.SVRKJSL in (.,.M) then &_V.OAGRDR = .M ;
else if &_V.SVXRRKR=1 then &_V.OAGRDR = .P ;
else if &_V.SVRKOST=.A or &_V.SVRKJSM=.A or &_V.SVRKJSL=.A then &_V.OAGRDR = .A ;

if &_V.SVLKOST=0 and &_V.SVLKJSM=0 and &_V.SVLKJSL=0 then &_V.OAGRDL = 0 ;
else if &_V.SVLKOST=1 and &_V.SVLKJSM=0 and &_V.SVLKJSL=0 then &_V.OAGRDL = 1 ;
else if &_V.SVLKOST=2 and &_V.SVLKJSM=0 and &_V.SVLKJSL=0 then &_V.OAGRDL = 2 ;
else if &_V.SVLKOST=2 and &_V.SVLKJSM in (0,1) and
    &_V.SVLKJSL in (0,1) and
    not (&_V.SVLKJSM=0 and &_V.SVLKJSL=0) then &_V.OAGRDL = 3 ;
else if &_V.SVLKOST=2 and (&_V.SVLKJSM=2 or &_V.SVLKJSL=2) then &_V.OAGRDL = 4 ;
else if &_V.SVLKOST=0 and &_V.SVLKJSM in (0,1) and
    &_V.SVLKJSL in (0,1) and
    not (&_V.SVLKJSM=0 and &_V.SVLKJSL=0) then &_V.OAGRDL = 1 ;

```

```

      not (&_V.SVLKJSM=0 and &_V.SVLKJSL=0)                                then &_V.OAGRDL = 1 ;
else if &_V.SVLKOST=1 and &_V.SVLKJSM in (0,1) and
      &_V.SVLKJSL in (0,1) and
      not (&_V.SVLKJSM=0 and &_V.SVLKJSL=0)                                then &_V.OAGRDL = 1 ;
else if &_V.SVLKOST=1 and (&_V.SVLKJSM=2 or &_V.SVLKJSL=2) then &_V.OAGRDL = 2 ;
else if &_V.SVLKOST=0 and (&_V.SVLKJSM=2 or &_V.SVLKJSL=2) then &_V.OAGRDL = 2 ;
else if &_V.SVLKOST in (.,.M) or &_V.SVLKJSM in (.,.M) or
      &_V.SVLKJSL in (.,.M)                                                then &_V.OAGRDL = .M ;
else if &_V.SVXRLKR=1                                                    then &_V.OAGRDL = .P ;
else if &_V.SVLKOST=.A or &_V.SVLKJSM=.A or &_V.SVLKJSL=.A then &_V.OAGRDL = .A ;

```

```

*****;
* Study: OAI *;
* Source name: ELGRISK.sas *;
* Source type: Program *;
* Author: Fran Harris *;
* Created: 04/12/2007 *;
* Input: DAF.IEI (Initial Eligibility Interview) *;
*       DAF.SV (Screening Visit) *;
*       DAF.EV (Enrollment Visit) *;
* Output: Variable: P02ELGRISK *;
*         Temporary dataset: TempRisk *;
* Summary: Indicates whether a participant had knee symptoms *;
*          with no risk factors, or risk factors without knee *;
*          symptoms, or both, or neither. *;
*          Possible values are: *;
*          0: Control cohort (no knee sx, no risk factors) *;
*          1: Incidence cohort: Knee symptoms only *;
*          2: Incidence cohort: Risk factors only *;
*          3: Incidence cohort: Both knee sx & risk factors *;
*          4: Incidence cohort: By protocol exception *;
*          5: Progression cohort (had symptomatic knee OA) *;
* Revisions: None to date *;
*****;
data TempRisk(keep=id P02ELGRISK);
  merge daf.sv(keep=id p01ksx p01kpmedcv)
        daf.iei(keep=id p02wtga p02kinj p02ksurg p02famhxkr p02jbmpcv
                    p02ikprisk p02actrisk)
        daf.ev(keep=id v00cohort v00age);
  by id;

  if v00cohort=1 then P02ELGRISK=5; /*Non-eligible control cohort*/
  else
  if v00cohort=3 then P02ELGRISK=0; /*Progression cohort*/
  else
  if v00cohort=2 and id in ('4600116','4600253','4600012','4601408','3501494','2301934','580
3466','4601713','2302820')
  then P02ELGRISK=4; /*Cohort assigned by protocol exception */
  else
  if v00cohort=2 and 45<=v00age<50 then do;
    if (p01ksx in (3,4,5) or p02ikprisk in (1,2) or p01kpmedcv=1) and
      (p02wtga=1 or p02kinj=1 or p02ksurg=1 or p02famhxkr=1 or p02jbmpcv=3 or p02actrisk=1
)
      then P02ELGRISK=3;
  end;
  else if v00cohort=2 and 50<=v00age<=79 then do;
    if (p01ksx in (3,4,5) or p01kpmedcv=1) and
      (p02wtga ne 1 and p02kinj ne 1 and p02ksurg ne 1 and p02famhxkr ne 1 and p02jbmpc
v ne 3)
      then P02ELGRISK=1;
    else if (p01ksx not in (3,4,5) and p01kpmedcv ne 1) and
      (p02wtga=1 or p02kinj=1 or p02ksurg=1 or p02famhxkr=1 or p02jbmpcv=3 or
p02ikprisk in (1,2) or p02actrisk=1)
      then P02ELGRISK=2;
    else if (p01ksx in (3,4,5) or p01kpmedcv=1) and
      (p02wtga=1 or p02kinj=1 or p02ksurg=1 or p02famhxkr=1 or p02jbmpcv=3)
      then P02ELGRISK=3;
  end;
run;

```

```
*****;
* Study: OAI *;
* Source name: EmployG.sas *;
* Source type: Program *;
* Author: Emily Scott *;
* Created: 09/14/2006 *;
* Input: &_lib..&_Mmo (Clinic Visit Workbook) *;
* Input variables are: &_v.CUREMP, &_v.UNPAYWK and *;
* &_v.NWKHLTH *;
* Output: Variable &_v.CEMPLOY *;
* Summary: Calculate current employment status from *;
* Clinic Visit Workbook Q45(p24) (Baseline), *;
* Q79(p40Q) (Follow-up) *;
* Revisions: Updated code so that assignment of values and *;
* special missing codes are handled properly, no *;
* released data were affected by incorrect *;
* code previously used. Corrected header and *;
* comments to include baseline. ES 03/13/07 *;
*****;
```

This code is written for use with data from any dataset,
so it uses generic macro variables to take the place of variable prefixes
that change from year to year, as follows:

&_v. stands for the variable prefixes V00, V01, V02, etc.

****;

1=Works for pay
2=Unpaid work for family business
3=Not working for health reasons
4=Not working other reasons
.D=Dont Know
.R=Refused
.M=Missing

Participant "Works for pay" if &_v.CUREMP=1 (YES)

Participant does "Unpaid work" if &_v.CUREMP=0 (NO) or .D (DK) and &_v.UNPAYWK=1 (YES)

Participant is "Not working/health reasons" if &_v.CUREMP=0 (NO) or .D (DK) and &_v.NWKHLTH=1 (YES)

Participant is "Not working/other reasons" if &_v.CUREMP=0 (NO) or .D (DK) and &_v.NWKHLTH=0 (NO)

Question was "Refused" if &_v.CUREMP=.R (Refused)

Question is "Dont Know" if &_v.UNPAYWK=.D (DK) and &_v.NWKHLTH is unanswered OR
if &_v.UNPAYWK=0 (NO) or .D (DK) and &_v.NWKHLTH=.D (DK)

****;

```
if &_v.CUREMP=1 then &_v.CEMPLOY=1;
else if &_v.CUREMP=.R then &_v.CEMPLOY=&_v.CUREMP;
else if &_v.CUREMP IN(0,.D) then do;
  if &_v.UNPAYWK=1 then &_v.CEMPLOY=2;
  else if &_v.NWKHLTH=1 then &_v.CEMPLOY=3;
  else if &_v.NWKHLTH=0 then &_v.CEMPLOY=4;
  else if ((&_v.UNPAYWK=.D) or (&_v.NWKHLTH=.D)) then &_v.CEMPLOY=.D;
  else if ((&_v.UNPAYWK<=.z) or (&_v.NWKHLTH<=.z)) then &_v.CEMPLOY=.M;
end;
else if &_v.CUREMP<=.z then &_v.CEMPLOY=&_v.CUREMP;
```

```

*****;
* Study: OAI *;
* Source name: EnrExcpt.sas *;
* Source type: Program *;
* Author: Emily Kenyon *;
* Created: 1/7/06 *;
* Input: daf.EnrollmentException (Enrollment Exception Notification form) *;
* *;
* daf.ph (Participant characteristics file) *;
* Output: work.EEexcept *;
* Summary: Create indicator variable for existence of Enrollment Exception *;
* Notification for participant. This form was required if the enrollment *;
* did not go entirely smoothly and they either did not meet the minium *;
* requirements of MRI, SV x-ray, and blood collection, or they did not *;
* meet the enrollment criteria for the cohort they ended up in. *;
* *;
* Revisions: None to date *;
*****;

data EEexcept (KEEP=ID V00ENREXCP);
    merge daf.EnrollmentException (in=inben)
/*create this variable for all those who were ultimately enrolled*/
    daf.ph (in=inp where=(enroll=1) keep=id enroll);
    by id; if inp;
*If in the enrollment exception dataset, set the indicator to 1;
    if inben then V00ENREXCP=1;
*Otherwise, indicator is set to 0;
    else V00ENREXCP=0;
run;

```

```

*****;
* Study: OAI *;
* Source name: Handed.sas *;
* Source type: Program *;
* Author: Emily Kenyon *;
* Created: 1/6/06 *;
* Input: daf.evhandxr (EV hand x-ray tracking form ) *;
* Output: work.Handed *;
* Summary: Calculates handedness from first non-missing hand x-ray tracking form. *;
* Revisions: Changed to use only dominant hand question 06/14/06 EK *;
* Updated to accommodate new special missing for Don't Know (.D) 03/23/07 EK *;
* Updated to include participants whose handed question is missing 5/1/07 EK *;
*****;
*Sort by id and handedness question in case on is missing the handedness value;
proc sort data=daf.evhandxr out=tempek;
by id v00rlhand;
run;
*And now take last value, which will be the non-missing value if there is one (checked
separately for non-agreement between first/repeat hand xray tracking forms. There were none;
data tempek;
    set tempek;
    by id;
if last.id;
run;
proc sort;by id;run;
data handed (keep=id V00HANDED);
    set tempek;by id;
    if first.id;
    V00HANDED=v00rlhand;
run;
*Clean-up;
proc datasets library=work memtype=data;
    delete tempek;
quit;

```

```

*****;
* Study: OAI *;
* Source name: IKPRISK.sas *;
* Source type: Program *;
* Author: Emily Kenyon *;
* Created: 04/06/2007 *;
* Input: daf.IEI (Initial Eligibility Interview) *;
* Output: P02IKPRISK *;
* Summary: Determines whether participant had infrequent *;
* (but not frequent) knee pain at the Initial *;
* Eligibility Interview. Was not used as a risk *;
* factor for eligibility at first, later added as a *;
* qualifying knee symptom for 45-49 years olds, and *;
* as a qualifying risk factor for 50-79 year olds *;
* *;
* Revisions: None to date *;
*****;

**** 0 = No knee pain either knee at IEI
      1 = Infrequent knee pain only at IEI
        2 = Frequent pain at least one knee at IEI
        .D = Don't know
        .R = Refused
      .M = Missing
      .A = Not expected
****;

      if p02kpn=0 then P02IKPRISK=0;
      else if p02kpn=1 and p02kpnr ne 1 and p02kpn1 ne 1 then P02IKPRISK=1;
      else if p02kpn=1 and (p02kpnr=1 or p02kpn1=1) then P02IKPRISK=2;
      else if p02kpn=.D and p02kpnr ne 1 and p02kpn1 ne 1 then P02IKPRISK=.D;
      else if p02kpn=.R and p02kpnr ne 1 and p02kpn1 ne 1 then P02IKPRISK=.R;
      else if p02kpn=.M or P02kpnr=.M or p02kpn1=.M then P02IKPRISK=.M;
      else if p02kpn=.A then P02IKPRISK=.A;

```

```

*****;
* Study: OAI *;
* Source name: IncomeG.sas *;
* Source type: Program *;
* Author: Emily Scott *;
* Created: January 16, 2006 *;
* Input: &_lib..&_Mmo (Self-administered questionnaire) *;
* Output: Variables &_v.INCOME and &_v.INCOME2 *;
* Summary: &_v.INCOME: Calculate categories of family income based on *;
* based on answers to &_v.INC10, &_v.INC25, &_v.INC50, and *;
* &_v.INC100. If a category cannot be determined due to *;
* missing data, the calculated variable &_v.INCOME is *;
* set to .R:Refused if any of the input questions were *;
* refused, or set to .D:Dont know if any of the input *;
* questions were answered as Dont Know. If data is *;
* missing for any other reason, the value is set to .M *;
* &_v.INCOME2: Calculate categories of family income based *;
* on answers to &_v.INC10, &_v.INC25, &_v.INC50, &_v.INC100 *;
* and &_v.INCOME. If a category cannot be determined due to *;
* missing data, calculated variable is set to a missing *;
* code. *;
* Revisions: [Alert 0.2.1] *;
* Baseline handling of refused changed from 77 to .R *;
* don't know from 88 to .D to match latest conventions; *;
* some baseline values for &_v.INCOME have changed as a *;
* result. *;
* Comments and header updated to apply to all visits. *;
* Added code to create new calculated variable not *;
* included at first baseline release - &_v.INCOME2. *;
* ES 03/23/07 *;
*****;

*This code is written for use with data from any dataset,
so it uses generic macro variables to take the place of variable prefixes
that change from year to year, as follows:
    &_v. stands for the variable prefixes P02, P01, V00, V01, V02, etc.*;

if &_v.inc25=0 and &_v.inc10=0 then &_v.INCOME=1; else /* <$10K */
if &_v.inc25=0 and &_v.inc10=1 then &_v.INCOME=2; else /* $10K-25K */
if &_v.inc25=1 and &_v.inc50=0 then &_v.INCOME=3; else /* $25K-50K */
if &_v.inc50=1 and &_v.inc100=0 then &_v.INCOME=4; else /* $50K-100K */
if &_v.inc50=1 and &_v.inc100=1 then &_v.INCOME=5; else /* >$100K */
                                &_v.INCOME=.M; /* Missing */

if &_v.INCOME not in (1,2,3,4,5) then do;
    if &_v.inc10=.R or &_v.inc25=.R or &_v.inc50=.R or &_v.inc100=.R then &_v.INCOME=.R; /*
Refused */
    else
        if &_v.inc10=.D or &_v.inc25=.D or &_v.inc50=.D or &_v.inc100=.D then &_v.INCOME=.D; /*
Dont know */
end;

*&_v.INCOME sets those that partially completed the income items to a SAS
special missing value code. &_v.INCOME2 captures some of these participants and
classifies them as >$50K or <$50K;
if &_v.INCOME in (1,2,3) then &_v.INCOME2=1; else /* <$50K */
if &_v.INCOME in (4,5) then &_v.INCOME2=2; else /* >$50K */
if &_v.INCOME<0 then do;
    if &_v.inc25=0 and &_v.inc10<0 then &_v.INCOME2=1; else /* <$50K */
    if &_v.inc25=1 and &_v.inc50=1 then &_v.INCOME2=2; else /* >$50K */
    if &_v.inc100=1 then &_v.INCOME2=2; else /* >$50K */
    if &_v.INCOME in (.M,.R,.D) then &_v.INCOME2=&_v.INCOME; /* Missing, refused or don'
t know */
end;

```

```

*****;
* Study: OAI
* Source name: JBMPCV.sas
* Source type: Program
* Author: Laura Akin
* Created: 03/31/2005
* Input: DAF.IEI (Initial Eligibility Interview)
* Output: &_v.JBMPCV:Hard bumps on joints closest to fingertips
* Summary: Calculating Hard bumps on joints
*          from IEI Q16,16a on page 6
* Revisions: Updated to use macro variables MD 04/02/07
*            Simplified code MD 06/04/07
*****;
*This code is written for use with data from any dataset,
so it uses generic macro variables to take the place of variable prefixes
that change from year to year, as follows:
    &_v. stands for the variable prefix P02*;

*** 0=No bumps
    1=Yes, right hand
    2=Yes, left hand
    3=Yes, both hands
***;
* if no hard bumps seen, set &_v.JBMPCV to 0;
  if &_v.JBMP=0 then &_v.JBMPCV=0;
* if main var is missing, set &_v.JBMPCV to missing;
  else if &_v.JBMP<=.z then &_v.JBMPCV=&_v.JBMP;
* if hard bumps seen, then &_v.JBMPCV indicates which hand has bumps;
  else if &_v.JBMP=1 then &_v.JBMPCV=&_v.JBMP2;

```

```

*****;
* Study: OAI *;
* Source name: KRS3CV.sas *;
* Source type: Program *;
* Author: Laura Akin *;
* Created: 03/31/2005 *;
* Input: DAF.IEI (Initial Eligibility Interview) *;
* Output: &_v.KRS3CV: Planned Knee Replacement Surgery *;
* Summary: Calculating Planned Knee surgery/knee replacement *;
* from IEI Q13,13a on page 5 *;
* Revisions: Updated to use macroitized variables MD 06/01/07 *;
*****;
* This code is written for use with data from any dataset, so it uses
  generic macro variables to take the place of variable prefixes that change from year
  to year, as follows:
    &_v. stands for the variable prefix P02 ;

*** 0=No planned knee replacement surgery, 1=Yes, right knee, 2=Yes, left knee 3=Yes, bo
th knees **;
* Check response to main knee replacement surgery question;
  * If not planning knee replacement surgery, set CV to 0;
if &_v.KRS3=0 then &_v.KRS3CV=0;
  * If planning knee replacement surgery question is missing, set CV to missing;
else if &_v.KRS3<=.z then &_v.KRS3CV=&_v.KRS3;
  * If planning knee replacement surgery;
else if &_v.KRS3=1 then do;
  * assign value depending on which knee(s) they're planning to have replaced;
  if &_v.KRPS3=1 then &_v.KRS3CV=1;
  else if &_v.KRPS3=2 then &_v.KRS3CV=2;
  else if &_v.KRPS3=3 then &_v.KRS3CV=3;
end;

```

```

/*****
**
*       Study: OAI
* Source name: LIVENOG.sas
* Source type: Program
*       Author: Maurice Dockrell
*       Created: 05/25/2006
*       Input:  &_lib.._Mmo (Clinic Visit Workbook)
*       Output: &_v.LIVENO
*       Summary: Creates the variable &_v.LIVENO, which collapses &_v.LIVENUM values greater t
han 5
*               to preserve confidentiality. Q4 p3 (Baseline SAQ), Q4d p3a (Follow-up SAQ)
* Revisions: Updated to run for all years MD 06/01/07
* [Alert 0.2.1] Values of &_v.LIVENO that were previously coded as .N are now assigned
* the special missing value that is assigned to &_v.LIVEALO
* [Alert 0.2.1] 07/25/07 Maximum group changed from "more than 6" to "6 or more" (inclusiv
e)
* [Alert 0.2.1] 07/25/07 Previous logic would only set &_v.LIVENO=0 if &_v.LIVENUM=0, howe
ver
*       there are cases where &_v.LIVENUM is left blank when participant lives alone, so
*       added logic that will use &_v.LIVEALO when &_v.LIVENUM is missing.
*****/
* This code is written for use with data from any dataset, so it uses
generic macro variables to take the place of variable prefixes that change from year
to year, as follows:
    &_v. stands for the variable prefixes P01, V01, V02, etc;

* if &_v.LIVENUM is less than 6 then set CV to that value;
if 0 <= &_v.LIVENUM <= 5 then &_v.LIVENO=&_v.LIVENUM;
* if &_v.LIVENUM is 6 or more then set CV = 6 ;
else if &_v.LIVENUM > 5 then &_v.LIVENO=6;
* if no answer for &_v.LIVENUM and &_v.LIVEALO=1 indicating person ;
* lives alone, then set &_v.LIVENO=0;
else if &_v.LIVEALO = 1 then &_v.LIVENO = 0 ;
* else set CV to missing value assigned to &_v.LIVEALO;
else if &_v.LIVEALO<=.z then &_v.LIVENO=&_v.LIVEALO;

```

```

/*****
**
*      Study: OAI
* Source name: LTHS.sas
* Source type: Program
*      Author: Maurice Dockrell
*      Created: 05/22/2006
*      Input:  &_lib.._Mmo: (Enrollment Visit SAQ)
*      Output: &_v.EDCV.
*      Summary: Modifies &_v.EDUCAT (Q6 p3) to combine all grades below 'high school graduate
',
               to preserve confidentiality.
*      Revisions: Updated to use macroized variables MD 04/10/07
*
*****/
* This code is written for use with data from any dataset, so it uses
generic macro variables to take the place of variable prefixes that change from year
to year, as follows:
    &_v. stands for the variable prefixes V00, V01, V02, etc;

* Collapse 'less than high school graduate' values;

if &_v.EDUCAT in (1,2,3,4) then &_v.EDCV=0;
  else if &_v.EDUCAT=5 then &_v.EDCV=1;
  else if &_v.EDUCAT=6 then &_v.EDCV=2;
  else if &_v.EDUCAT=7 then &_v.EDCV=3;
  else if &_v.EDUCAT=8 then &_v.EDCV=4;
  else if &_v.EDUCAT=9 then &_v.EDCV=5;
  else &_v.EDCV=&_v.EDUCAT;

```

```

*****;
* Study: OAI
* Source name: MOMHRtoBROHRCVG.sas
* Source type: Program
* Author: Laura Akin
* Created: 01/31/2005
* Input: &_lib..&_Mmo (Clinic Visit Workbook)
* Output: &_v.MOMHRCV: History of hip replacement: Mother
*         &_v.DADHRCV: History of hip replacement: Father
*         &_v.SISHRCV: History of hip replacement: Sister
*         &_v.BROHRCV: History of hip replacement: Brother
*
* Summary: Calculating Family History of Hip Replacement
*          Q12,12a-d on page 5 (Baseline) Q4,4a-d page 3Q (Follow-up)
* Revisions: Updated to run for all years MD 04/01/07
*****;
* This code is written for use with data from any dataset, so it uses
  generic macro variables to take the place of variable prefixes that change from year
  to year, as follows:
    &_v. stands for the variable prefixes P01, V01, V02, etc;
***
  Bringing NO/DK/REF down from main variable
***;

%collaps2(&_v.FAMHR,&_v.MOMHR,&_v.MOMHRCV);
%collaps2(&_v.FAMHR,&_v.DADHR,&_v.DADHRCV);
%collaps2(&_v.FAMHR,&_v.SISHR,&_v.SISHRCV);
%collaps2(&_v.FAMHR,&_v.BROHR,&_v.BROHRCV);

```

```

*****;
* Study: OAI
* Source name: MOMKRtoBROKRCVG.sas
* Source type: Program
* Author: Laura Akin
* Created: 01/31/2005
* Input: &_lib..&_Mmo (Clinic Visit Workbook)
* Output: &_v.MOMKRCV: History of knee replacement: Mother
*         &_v.DADKRCV: History of knee replacement: Father
*         &_v.SISKRCV: History of knee replacement: Sister
*         &_v.BROKRCV: History of knee replacement: Brother
*
* Summary: Calculating Family History of Knee Replacement
*          Q13,13a-d on page 5 (Baseline) Q5,5a-d page 3Q (Follow-up)

* Revisions: Updated to run for all years MD 04/01/07
*****;
* This code is written for use with data from any dataset, so it uses
  generic macro variables to take the place of variable prefixes that change from year
  to year, as follows:
    &_v. stands for the variable prefixes P01, V01, V02, etc;

***
  Bringing NO/DK/REF down from main variable
***;

%collaps2(&_v.FAMKR,&_v.MOMKR,&_v.MOMKRCV);
%collaps2(&_v.FAMKR,&_v.DADKR,&_v.DADKRCV);
%collaps2(&_v.FAMKR,&_v.SISKR,&_v.SISKRCV);
%collaps2(&_v.FAMKR,&_v.BROKR,&_v.BROKRCV);

```

```

*****;
* Study: OAI
* Source name: MRKCBCVG.sas
* Source type: Program
* Author: Laura Akin
* Created: 01/31/2005
* Input: &_lib.._Mmo (Clinic Visit)
* Output: &_v.MRKBCBV: Participant's knee fits comfortably in coil
*
* Summary: Calculating knee replacement/coil fit from
*           SVW Q1a-c on page 30
*           &_v.MRKBCBV is assigned a value to indicate which knee(s)
*           fit inside the coil and whether or not there is a
*           knee replacement
* Revisions: Updated to run for all years, including baseline MD 04/03/07
*           [Alert 0.2.1] format description corrected. Descriptions for values of
*           &_v.MRKBCBV =4 and 5 reversed. Affects releases 0.2.1 MD 06/04/07
*****;
*This code is written for use with data from any dataset,
so it uses generic macro variables to take the place of variable prefixes
that change from year to year, as follows:
    &_v. stands for the variable prefixes P01, V01, V02, etc.*;

***
0=No
1=Yes, right knee only
2=Yes, left knee only
3=Yes, both knees
4=Yes, left knee only, right knee Total Knee Replacement
5=Yes, right knee only, left knee Total Knee Replacement
***;

* if no knee replacement surgery, set &_v.MRKBCBV to &_v.MRKCB;
if &_v.MRSCAN=0 and &_v.MRKCB in (1,2,3,0) then &_v.MRKBCBV=&_v.MRKCB;
* if right knee replaced and left knee fits in coil;
  else if &_v.MRSCAN=1 and &_v.MRKCL=1 then &_v.MRKBCBV=4;
* if left knee replaced and right knee fits in coil;
  else if &_v.MRSCAN=2 and &_v.MRKCR=1 then &_v.MRKBCBV=5;
* assign special missing values;
  else if &_v.MRSCAN<=.z then &_v.MRKBCBV=&_v.MRSCAN;
  else if &_v.MRKCB<=.z then &_v.MRKBCBV=&_v.MRKCB;

```

```

*****;
* Study: OAI
* Source name: MRPRB.sas
* Source type: Program
* Author: Laura Akin
* Created: 01/06/2006
* Input: daf.SV (Screening Visit)
* Output: &_v.MRPRBCV
* Summary: Calculating problems related to MRI scan
*          Combining SV MRI Screening Interview Q1,Q1a (p25)
*
* Revisions: Updated to use macro variables MD 04/02/07
*****;
* This code is written for use with data from any dataset, so
* it uses generic macro variables to take the place of library,
* dataset prefixes and suffixes, and variable prefixes that change from year
* to year, as follows:
*     &_v. stands for the variable prefix P01.

***
    Bringing NO/DK/REF down from main variable
***;

%collaps2(&_v.MRIB4,&_v.MRIPROB,&_v.MRPRBCV);

```

```

*****;
* Study: OAI
* Source name: PA130CVto530CVG.sas
* Source type: Program
* Author: Maurice Dockrell
* Created: 09/25/2006
* Input: &_lib.&_Mmo (Clinic Visit workbook)
* Output:  &_v.PA230CV:   How often kneeling for 30 min during single day
*          &_v.PA330CV:   How often squatting for 30 min during single day
*          &_v.PA530CV:   How often in/out of squatting position 10+ time/day
*          &_v.PA430CV:   How often lifting 25lbs in a single day
*          &_v.PA130CV:   How often climbing 10+ flights of stairs/day
*
* Summary:  Calculating frequency of knee bending activities from Clinic
*           Visit workbook p16 (Baseline) p32Q (Follow-up) (&_v.PA230 and &_v.PA230FQ, kneel
ing),
*           (&_v.PA330 and &_v.PA330FQ, squatting),
*           Visit workbook p17 (Baseline) p33Q (Follow-up) (&_v.PA530 and &_v.PA530FQ, in an
d out of squat 10x),
*           (&_v.PA430 and &_v.PA430FQ, lift/move 25 lb,
*           Visit workbook p18 (Baseline) p34Q (Follow-up) (&_v.PA130 and &_v.PA130FQ, stair
s)
* Revisions: Updated to run for all years, including baseline MD 03/16/07

*****;
* This code is written for use with data from any dataset, so
* it uses generic macro variables to take the place of library,
*
* dataset prefixes and suffixes, and variable prefixes that change from year
* to year, as follows:
*   &_lib. stands for the SAS library containing the data for the year being
*         processed (e.g., daf, daf1, daf2, etc)
*   &_Mmo. stands for the visit prefix for the dataset (e.g., EV, M12, M24, etc)
*   &_v.  stands for the variable prefixes V00, V01, V02, etc.

*** 0=None, 1=One day or less/wk, 2=2-3/wk, 3=4-5/wk 4=Nearly every day/every day   **;

%collaps2(&_v.PA230,&_v.PA230FQ,&_v.PA230CV);
%collaps2(&_v.PA330,&_v.PA330FQ,&_v.PA330CV);
%collaps2(&_v.PA530,&_v.PA530FQ,&_v.PA530CV);
%collaps2(&_v.PA430,&_v.PA430FQ,&_v.PA430CV);
%collaps2(&_v.PA130,&_v.PA130FQ,&_v.PA130CV);

```

```

*****;
* Study: OAI *;
* Source name: PASEG.sas *;
* Source type: Program *;
* Author: Emily Scott *;
* Created: 09/15/06 *;
* Input: &_lib.&_Mmo (Clinic Visit Workbook) *;
* Input variables are: *;
* &_v.PASE2, &_v.PASE2HR, &_v.PASE3, &_v.PASE3HR, &_v.PASE4, &_v.PASE4HR, *;
* &_v.PASE5, &_v.PASE5HR, &_v.PASE6, &_v.PASE6HR, &_v.HOUACT1, &_v.HOUACT2, *;
* &_v.HOUACT3, &_v.HOUACT4, &_v.HOUACT5, &_v.HOUACT6, &_v.WORK7, *;
* &_v.WORKHR7, &_v.WORK1HR, and &_v.WORKAMT. *;
* Output: Temporary dataset: work.finalPase *;
* Variable: &_v.PASE *;
* Summary: Calculate the Physical Activity Scale for the Elderly (PASE). *;
* *;
* A series of twelve questions (some with subquestions regarding frequency *;
* per week, hours per day, and hours per week) are asked about physical *;
* activity levels in the last seven days. Resulting score is rounded to the *;
* nearest integer, with a theoretical range from 0 to over 400. Higher *;
* scores indicate greater level of physical activity. If any of the *;
* required 12 questions or subquestions are missing, then the calculated *;
* score is set to a special missing value. *;
* *;
* - Frequency of the activity is collected as: *;
* (a) days per week (never, 1-2 days, 3-4 days, or 5-7 days), and *;
* (b) the number of hours per day (<1, 1 to <2, 2 to 4, more than 4), or *;
* (c) yes/no (1, 0), or *;
* (d) hours worked per week (0 to 144). *;
* - Questions of type (a) are translated as the midpoints of the frequency *;
* range (i.e., 0, 1.5, 3.5, or 6, respectively, for days of week) and *;
* type (b) are translated to the midpoints of the hours range (i.e., 0.5, *;
* 1.5, 3, or 5, respectively) are used to calculate a subscale of hours *;
* per day (freq*hrs/7). Twelve subscales are weighted depending on the *;
* strenuousness of the activity, and then summed to give an overall PASE *;
* score. *;
* *;
* Revisions: [Alert 0.2.1 and 1.1.2] *;
* Some baseline and 12-month values have changed as a result of chang
ing *;
* how missings are handled. Code was updated so that if hours per day items *;
* and/or the occupational activity item are missing, then the median of the *;
* item is obtained using all available data for that visit. This median is *;
* used in place of the missing data. If the obtained median is a decimal *;
* number, it is rounded down. If there are more than 3 hours per *;
* day/occupational activity level items missing, then the score is set to *;
* missing. If any of the activity frequency items, household items or key *;
* work items are missing, the score is set to missing. Also, code was *;
* updated to set the PASE score to more tailored special missing values, *;
* depending on the missing values assigned to the raw variables. ES 04/11/07*
*****;

*This code is written for use with data from any dataset, so it uses
generic macro variables to take the place of library, dataset prefixes and suffixes,
and variable prefixes that change from year to year, as follows:
    &_lib. stands for the SAS library containing the data for the year being processed (
e.g., daf, daf1, daf2, etc.)
    &_Mmo. stands for the visit prefix for the dataset (e.g., EV, M12, M24, etc.)
    &_v. stands for the variable prefixes V00, V01, V02, etc.*;

%Macro items16(often, hours, OFTEN1, HOURS1, MED, FREQUEN);
***RECODE &OFTEN TO &OFTEN1 FREQ***;
IF &OFTEN=0 THEN &OFTEN1=0;
IF &OFTEN=1 THEN &OFTEN1=1.5;
IF &OFTEN=2 THEN &OFTEN1=3.5;
IF &OFTEN=3 THEN &OFTEN1=6;
***RECODE &HOURS TO &HOURS1***;
IF &HOURS<0 AND &OFTEN1 IN (1.5, 3.5, 6) THEN &HOURS1=&MED;

```

```

IF &HOURS=1 THEN &HOURS1=0.5;
IF &HOURS=2 THEN &HOURS1=1.5;
IF &HOURS=3 THEN &HOURS1=3;
IF &HOURS=4 THEN &HOURS1=5;
IF &HOURS=. THEN &HOURS1=0;
***DETERMINE ACTIVITY FREQUENCY***;
&FREQUEN=(&OFTEN1*&HOURS1)/7;
%MEND ITEMS16;

***GET RAW DATA AND TRANSFORM &_v.WORKAMT INTO DICHOTOMOUS VARIABLE***;
data getDataPase (keep= ID &_v.PASE1 &_v.PASE2 &_v.PASE2HR &_v.PASE3 &_v.PASE3HR &_v.PASE4 &
_v.PASE4HR
&_v.PASE5 &_v.PASE5HR &_v.PASE6 &_v.PASE6HR &_v.HOUACT1
&_v.HOUACT2
&_v.HOUACT3 &_v.HOUACT4 &_v.HOUACT5 &_v.HOUACT6 &_v.
WORK7
&_v.WORKHR7 &_v.WORK1HR &_v.WORKAMT &_v.WORKAMTX);
set &_lib..&_Mmo;
if &_v.WORKAMT=1 then &_v.WORKAMTX=1;
else if &_v.WORKAMT in (2,3,4) then &_v.WORKAMTX=2;
run;

%macro univ1(var,out,med);
proc univariate data=getDataPase noprint;
var &var;
output out=&out median=&med;
run;
%mend univ1;

***GET THE MEDIAN FOR HOURS PER DAY ITEMS AND OCCUPATIONAL ACTIVITY LEVEL (TRANSFORMED VARIA
BLE)***;
%univ1(&_v.PASE2HR,PASE2HRoutput,&_v.PASE2HRmedian);
%univ1(&_v.PASE3HR,PASE3HRoutput,&_v.PASE3HRmedian);
%univ1(&_v.PASE4HR,PASE4HRoutput,&_v.PASE4HRmedian);
%univ1(&_v.PASE5HR,PASE5HRoutput,&_v.PASE5HRmedian);
%univ1(&_v.PASE6HR,PASE6HRoutput,&_v.PASE6HRmedian);
%univ1(&_v.WORKAMTX,WORKAMToutput,&_v.WORKAMTmedian);

***IF THE MEDIAN IS A DECIMAL NUMBER, ROUND DOWN TO THE INTEGAR***;
data mergeAllPase;
merge PASE2HRoutput
PASE3HRoutput
PASE4HRoutput
PASE5HRoutput
PASE6HRoutput
WORKAMToutput;
&_v.PASE2HRmed=INT(&_v.PASE2HRmedian);
&_v.PASE3HRmed=INT(&_v.PASE3HRmedian);
&_v.PASE4HRmed=INT(&_v.PASE4HRmedian);
&_v.PASE5HRmed=INT(&_v.PASE5HRmedian);
&_v.PASE6HRmed=INT(&_v.PASE6HRmedian);
&_v.WORKAMTmed=INT(&_v.WORKAMTmedian);
run;

data pase;
if &_N_ = 1 then set work.mergeAllPase;
set getDataPase;
run;

data finalPase (keep = ID &_v.PASE);
set work.pase;

S*** &_v.PASE2F *** **CALCULATED ACTIVITY FREQUENCY, WALKING***;
%ITEMS16 (OFTEN=&_v.PASE2,HOURS=&_v.PASE2HR,OFTEN1=&_v.PASE21,HOURS1=&_v.PASE2HR1,MED=&_v.PAS
E2HRmed,
FREQUEN=&_v.PASE2F);
E*** &_v.PASE2F ***;

```

```

*$$$ &_v.PASE3F *** ***CALCULATED ACTIVITY FREQUENCY, LITE SPORT/REC***;
%ITEMS16(&_v.PASE3,&_v.PASE3HR,&_v.PASE31,&_v.PASE3HR1,&_v.PASE3HRmed,&_v.PASE3F);
*E$$$ &_v.PASE3F ***;

*$$$ &_v.PASE4F *** ***CALCULATED ACTIVITY FREQUENCY, MODERATE SPORT/REC***;
%ITEMS16(&_v.PASE4,&_v.PASE4HR,&_v.PASE41,&_v.PASE4HR1,&_v.PASE4HRmed,&_v.PASE4F);
*E$$$ &_v.PASE4F ***;

*$$$ &_v.PASE5F *** ***CALCULATED ACTIVITY FREQUENCY, STRENUOUS SPORT/REC***;
%ITEMS16(&_v.PASE5,&_v.PASE5HR,&_v.PASE51,&_v.PASE5HR1,&_v.PASE5HRmed,&_v.PASE5F);
*E$$$ &_v.PASE5F ***;

*$$$ &_v.PASE6F *** ***CALCULATED ACTIVITY FREQUENCY, MUSCLE STRENGTH/ENDURANCE EXERCISES**
*;
%ITEMS16(OFTEN=&_v.PASE6,HOURS=&_v.PASE6HR,OFTEN1=&_v.PASE61,HOURS1=&_v.PASE6HR1,MED=&_v.PAS
E6HRmed,
FREQUEN=&_v.PASE6F);
*E$$$ &_v.PASE6F ***;

*$$$ &_v.PASE01 *** COMPONENT: Walk outside home (Weight=20)***;
&_v.PASE01=20*&_v.PASE2F;
*E$$$ &_v.PASE01 *** ;

*$$$ &_v.PASE02 *** COMPONENT: Light sport/rec activity (Weight=21)***;
&_v.PASE02=21*&_v.PASE3F;
*E$$$ &_v.PASE02 ***;

*$$$ &_v.PASE03 *** COMPONENT: Moderate sport/rec activity (Weight=23)***;
&_v.PASE03=23*&_v.PASE4F;
*E$$$ &_v.PASE03 ***;

*$$$ &_v.PASE04 *** COMPONENT: Strenuous sport/rec activity (Weight=23)***;
&_v.PASE04=23*&_v.PASE5F;
*E$$$ &_v.PASE04 ***;

*$$$ &_v.PASE05 *** COMPONENT: Muscle Str,Endurance activity (Weight=30)***;
&_v.PASE05=30*&_v.PASE6F;
*E$$$ &_v.PASE05 ***;

*$$$ &_v.PASE06 *** COMPONENT: Light housework (Weight=25)***;
&_v.PASE06=25*&_v.HOUACT1;
*E$$$ &_v.PASE06 ***;

*$$$ &_v.PASE07 *** COMPONENT: Heavy housework (Weight=25)***;
&_v.PASE07=25*&_v.HOUACT2;
*E$$$ &_v.PASE07***;

*$$$ &_v.PASE08 *** COMPONENT: Home Repairs (Weight=30)***;
&_v.PASE08=30*&_v.HOUACT3;
*E$$$ &_v.PASE08 ***;

*$$$ &_v.PASE09 *** COMPONENT: Lawn work/yard care (Weight=36)***;
&_v.PASE09=36*&_v.HOUACT4;
*E$$$ &_v.PASE09 ***;

*$$$ &_v.PASE10 *** COMPONENT: Outdoor Gardening (Weight=20)***;
&_v.PASE10=20*&_v.HOUACT5;
*E$$$ &_v.PASE10 ***;

*$$$ &_v.PASE11 *** COMPONENT: Caring for another person (Weight=35)***;
&_v.PASE11=35*&_v.HOUACT6;
*E$$$ &_v.PASE11 ***;

*$$$ &_v.PASE12 *** COMPONENT: Work for pay/volunteer (Weight=21)*** ;
&_v.PASE12 = .;

if (&_v.WORK7<0 or &_v.WORK1HR=.D or &_v.WORKAMT=.D) then WORKMISS=1;

```

```

if &_v.WORK1HR=1 THEN &_v.WORKHR7=1; ***Work less than 1 hour counted as 1 hour of work***;
if (&_v.WORK7=1 and &_v.WORKAMT<0) then &_v.WORKAMT=&_v.WORKAMTmed; ***If occupational activity
is missing, use the median of transformed variable***;
if &_v.WORK7=0 THEN &_v.WORK7F=0; ***If did not work in past 7 days, frequency is 0***;
if &_v.WORKAMT=1 THEN &_v.WORK7F=0; ***If occupational activity level is sitting, frequency
is 0***;
    else if (&_v.WORKAMT in (2,3,4)) THEN &_v.WORK7F=&_v.WORKHR7/7;

&_v.PASE12=21*&_v.WORK7F;
*E*** &_v.PASE12 ***;

*S*** &_v.PASE *** Calculate total PASE score, rounded to nearest integer***;
&_v.PASE=Round(SUM(&_v.PASE01,&_v.PASE02,&_v.PASE03,&_v.PASE04,&_v.PASE05,&_v.PASE06,&_v.PASE07,
    &_v.PASE08,&_v.PASE09,&_v.PASE10,&_v.PASE11,&_v.PASE12),1);
*E*** &_v.PASE***;

*S*** Set to special missing value if any of the below responses are missing***;
if &_v.PASE2=.R or &_v.PASE3=.R or &_v.PASE4=.R or &_v.PASE5=.R or &_v.PASE6=.R or
    &_v.HOUACT1=.R or &_v.HOUACT2=.R or &_v.HOUACT3=.R or &_v.HOUACT4=.R or &_v.HOUACT5=.R or
    &_v.HOUACT6=.R or
    &_v.WORK7=.R then &_v.PASE=.R; ***.R = Refused***;
else if &_v.PASE2=.D or &_v.PASE3=.D or &_v.PASE4=.D or
    &_v.PASE5=.D or &_v.PASE6=.D or &_v.HOUACT1=.D or &_v.HOUACT2=.D or
    &_v.HOUACT3=.D or &_v.HOUACT4=.D or &_v.HOUACT5=.D or &_v.HOUACT6=.D or &_v.WORK7=.D or &
    &_v.WORK1HR=.D
    then &_v.PASE=.D; ***.D = Don't know***;
else if &_v.PASE1=.A and &_v.PASE2=.A and &_v.PASE3=.A and &_v.PASE4=.A and &_v.PASE5=.A and
    &_v.PASE6=.A and &_v.HOUACT1=.A and
    &_v.HOUACT2=.A and &_v.HOUACT3=.A and &_v.HOUACT4=.A and &_v.HOUACT5=.A and &_v.HOUACT6=.
    A and &_v.WORK7=.A
    then &_v.PASE=.A; ***.A = Not expected***;
else if &_v.PASE2=. and &_v.PASE3=. and &_v.PASE4=. and &_v.PASE5=. and &_v.PASE6=. and &_v.
    HOUACT1=. and
    &_v.HOUACT2=. and &_v.HOUACT3=. and &_v.HOUACT4=. and &_v.HOUACT5=. and &_v.HOUACT6=. and
    &_v.WORK7=.
    then &_v.PASE=.; ***. = Missing Form/Incomplete Workbook***;
else if &_v.PASE2<0 or &_v.PASE3<0 or &_v.PASE4<0 or &_v.PASE5<0 or &_v.PASE6<0 or &_v.HOUAC
    T1<0 or &_v.HOUACT2<0 or
    &_v.HOUACT3<0 or &_v.HOUACT4<0 or &_v.HOUACT5<0 or &_v.HOUACT6<0 or &_v.WORK7<0
    then &_v.PASE=.M; ***.M = Missing***;
*E*** Set to special missing value if any of the below responses are missing***;

*S*** Set to special missing value if there are more than 3 missing of hours per day/occupat
ional activity level items***;
if &_v.PASE>.z then do;
    HOURMISS=0;
    if      ((&_v.PASE2 IN (1,2,3)) AND (&_v.PASE2HR not in (1,2,3,4))) then HOURMISS + 1;
    if      ((&_v.PASE3 IN (1,2,3)) AND (&_v.PASE3HR not in (1,2,3,4))) then HOURMISS + 1;
    if      ((&_v.PASE4 IN (1,2,3)) AND (&_v.PASE4HR not in (1,2,3,4))) then HOURMISS + 1;
    if      ((&_v.PASE5 IN (1,2,3)) AND (&_v.PASE5HR not in (1,2,3,4))) then HOURMISS + 1;
    if      ((&_v.PASE6 IN (1,2,3)) AND (&_v.PASE6HR not in (1,2,3,4))) then HOURMISS + 1;

    if sum(HOURMISS, WORKMISS) > 3
        then &_v.PASE=.I; ***.I = Inadequate data***;
end;
*E*** Set to special missing value if there are more than 3 missing of hours per day/occupat
ional activity level items***;
run;

***Clean-up***;
proc datasets library=work memtype=data;
    delete getDataPase
        PASE2HRoutput PASE3HRoutput PASE4HRoutput PASE5HRoutput PASE6HRoutput WOR
        KAMToutput
        mergeAllPase pase;
quit;

```



```

*****;
* Study: OAI *;
* Source name: RKSX_LKSXG.sas *;
* Source type: Program *;
* Author: Emily Scott *;
* Created: 09/20/2006 *;
* Input: &_lib.&_Mmo (Clinic Visit Workbook) *;
* Output: Variables &_v.RKSX and &_v.LKSX *;
* *;
* Summary: Knee symptom status (&_v.RKSX, &_v.LKSX) based on pain *;
* most days of a month in the past 12 months, from *;
* Q#15,16,18,19 (pp.6-9) (Baseline Screening Visit), *;
* Q#6-9(pp.4Q-5Q) (Follow-up). *;
* *;
* Input variables for right knee are &_v.KPNR12, *;
* &_v.KPNR, &_v.KPNR30 and &_v.KPNR30M. Input variables *;
* for left knee are &_v.KPNL12, &_v.KPNL, &_v.KPNL30 and *;
* &_v.KPNL30M. *;
* *;
* Possible values for calculated variables are: *;
* 0 No pain in past 12 months *;
* 1 Pain in past 12m, but not most days of a month *;
* 2 Pain most days of a month in past 12 months *;
* .D Don't know *;
* .R Refused *;
* .A Not expected *;
* .M Missing data *;
* *;
* Revisions: Corrected header and comments to include baseline. *;
* ES 03/20/07 *;

```

```

*****;

```

*This code is written for use with data from any dataset,
so it uses generic macro variables to take the place of variable prefixes
that change from year to year, as follows:

&_v. stands for the variable prefixes P02, P01, V00, V01, V02, etc.*;

```

*Right knee*;
  if (&_v.kpnr12=1 and &_v.kpnr=1) or
    (&_v.kpnr30=1 and &_v.kpnr30m=1) then &_v.RKSX=2; /*Pain most days of a month in p
ast 12m*/
  else
    if (&_v.kpnr12=1 and &_v.kpnr ne 1) or
      (&_v.kpnr30=1 and &_v.kpnr30m ne 1) then &_v.RKSX=1; /*Pain in past 12m, but not most
days of a month*/
    else
      if (&_v.kpnr12=0 and &_v.kpnr30 ne 1) then &_v.RKSX=0; /*No pain in past 12m*/
      else
        if (&_v.kpnr12=.D and &_v.kpnr30 in (0,.D)) then &_v.RKSX=.D; /*Don't know*/
        else
          if (&_v.kpnr12=.R or &_v.kpnr=.R or &_v.kpnr30=.R or &_v.kpnr30m=.R) then &_v.RKSX=.
R; /*Refused*/
          else
            if (&_v.kpnr12=.A and &_v.kpnr=.A and &_v.kpnr30=.A and &_v.kpnr30m=.A) then &_v.RKS
X=.A; /*Not expected*/
            else
              if &_v.kpnr12=.M then &_v.RKSX=.M; /*Missing*/

*Left knee*;
  if (&_v.kpnl12=1 and &_v.kpnl=1) or
    (&_v.kpnl30=1 and &_v.kpnl30m=1) then &_v.LKSX=2; /*Pain most days of a month in p
ast 12m*/
  else
    if (&_v.kpnl12=1 and &_v.kpnl ne 1) or
      (&_v.kpnl30=1 and &_v.kpnl30m ne 1) then &_v.LKSX=1; /*Pain in past 12m, but not most
days of a month*/
    else

```

```

        if (&_v.kpnl12=0 and &_v.kpnl30 ne 1) then &_v.LKSX=0; /*No pain in past 12m*/
    else
        if (&_v.kpnl12=.D and &_v.kpnl30 in (0,.D)) then &_v.LKSX=.D; /*Don't know*/
        else
            if (&_v.kpnl12=.R or &_v.kpnl=.R or &_v.kpnl30=.R or &_v.kpnl30m=.R) then &_v.LKSX=.
R; /*Refused*/
            else
                if (&_v.kpnl12=.A and &_v.kpnl=.A and &_v.kpnl30=.A and &_v.kpnl30m=.A) then &_v.LKS
X=.A; /*Not expected*/
            else
                if &_v.kpnl12=.M then &_v.LKSX=.M; /*Missing*/

```

```

*****;
* Study: OAI *;
* Source name: STMED.sas *;
* Source type: Program *;
* Author: Laura Akin *;
* Created: 01/08/2006 *;
* Input: daf.IEI (Initial Eligibility Interview) *;
* Output: &_v.STMEDCV *;
* Summary: Determining whether participant is taking *;
* medications as part of any other research study *;
* (combining IEI Q23,Q23a, p12) *;
* *;
* Revisions: Updated to use macro variables MD 04/02/07 *;
*****;
* This code is written for use with data from any dataset, so
* it uses generic macro variables to take the place of library,
* dataset prefixes and suffixes, and variable prefixes that change from year
* to year, as follows:
* &_v. stands for the variable prefixes P02

**** 0 Not participating in research study
      2 No, not told
      1 Yes, told

****;
if &_v.STMED=0 then &_v.STMEDCV=0;
else if &_v.STMED<=.z then &_v.STMEDCV=&_v.STMED;
else if &_v.STMED=1 then do;
  if &_v.STMDK=0 then &_v.STMEDCV=2;
  else if &_v.STMDK=1 then &_v.STMEDCV=1;
  else if &_v.STMDK<=.z then &_v.STMEDCV=&_v.STMDK;
end;

```

```

*****;
* Study: OAI *;
* Source name: SxOA.sas (will be %INCLUDEd into code FFxTrackG.sas) *;
* Source type: Program *;
* Author: Fran Harris *;
* Created: 04/04/2005 *;
* *;
* Input: DAF.SV (data analysis file for screening visit) *;
* Temp dataset SVXRD, created by code FFxTrackG.sas *;
* *;
* Output: SAS dataset: WORK.SXOA *;
* Variables: ID, P01LSXKOA, p01LSXKOA, P01RSXKOA, *;
* P01XRKOA, P01LXRKOA, P01RXRKOAA, *;
* P01RXRKOAA2, P01LXRKOAA2, P01RKXSX, P01LKXSX *;
* *;
* Summary: Creates the following variables by knee: *;
* *;
* - symptomatic knee OA (P01LSXKOA, P01RSXKOA) *;
* based on definite osteophytes on x-ray and pain most *;
* days of a month in the past 12 months (Input variables *;
* for left knee are P01SVLKOST, P01KPNL, P01KPNL12, and *;
* P01KPNL30M. Input variables for right knee are *;
* P01SVRKOST, P01KPNR, P01KPNR12, and P01KPNR30M.) *;
* Possible values for calculated variables are: *;
* 0 No symptomatic OA, this knee *;
* 1 Symptomatic OA, this knee *;
* .R Refused questions on symptoms *;
* .P Prosthetic, this knee (no x-ray) *;
* .M Missing for other reason *;
* *;
* - x-ray knee OA status (P01LXRKOA, P01RXRKOAA) *;
* based on definite osteophytes on x-ray (Input variable *;
* for left knee is P01SVLKOST. Input variable for right *;
* knee is P01SVRKOST.) *;
* Possible values for calculated variables are: *;
* 0 No x-ray OA, this knee *;
* 1 X-ray OA, this knee *;
* .P Prosthetic, this knee (no x-ray) *;
* .M Missing for other reason *;
* *;
* - detailed x-ray knee OA status (P01LXRKOAA2, P01RXRKOAA2) *;
* based on osteophytes (definite or possible), and on the *;
* maximum degree of medial/lateral joint space narrowing *;
* (none, mild-moderate, severe) (Input variables for left *;
* knee are P01SVLKOST, P01SVLKJSM, and P01SVLKJSL. Input *;
* variables for right knee are P01SVRKOST, P01SVRKJSM, *;
* and P01SVRKJSL.) *;
* Possible values for calculated variables are: *;
* 0 No osteophytes, no JSN *;
* 1 No osteophytes, definite JSN *;
* 2 Possible osteophytes, no JSN *;
* 3 Possible osteophytes, definite JSN *;
* 4 Definite osteophytes, no JSN *;
* 5 Definite osteophytes, mild/moderate JSN *;
* 6 Definite osteophytes, severe JSN *;
* .P Prosthetic, this knee (no x-ray) *;
* .M Missing for other reason *;
* *;
* - knee symptom status (P01LKXSX, P01RKXSX) based on pain *;
* most days of a month in the past 12 months. (Input *;
* variables for left knee are P01KPNL, P01KPNL12, *;
* P01KPNL30 and P01KPNL30M. Input variables for right *;
* knee are P01KPNR, P01KPNR12, P01KPNR30 and P01KPNR30M.) *;
* Possible values for calculated variables are: *;
* 0 No pain in past 12 months *;
* 1 Pain in past 12m, but not most days of a month *;
* 2 Pain most days of a month in past 12 months *;
* .R Refused to answer pain questions *;

```

```

*      .D Don't know *;
*      .A Not expected *;
*      .M Missing data, other than refusal to answer *;
*
* Creates the following variables by person: *;
*
* - symptomatic knee OA (P01SXKOA) *;
*   based on definite osteophytes on x-ray and pain most *;
*   days of a month in the past 12 months *;
*   Possible values for calculated variables are: *;
*     0 Neither knee *;
*     1 Right knee only *;
*     2 Left knee only *;
*     3 Both knees *;
*   .R Refused to answer symptom questions, both knees *;
*   .M Missing data for both knees *;
*
* - x-ray knee OA status (P01XRKOA) *;
*   based on definite osteophytes on x-ray *;
*   Possible values for calculated variables are: *;
*     0 Neither knee *;
*     1 Right knee only *;
*     2 Left knee only *;
*     3 Both knees *;
*   .M Missing data for both knees *;
*
* Calculated variables based on knee symptoms are set to *;
* .R:Refused if the questions regarding knee pain were *;
* refused. *;
* Calculated variables based on x-ray results are set to *;
* .P:Prosthesis if the knee has been replaced. *;
* Calculated variables are set to missing if any of the *;
* input variables are missing for any other reason. *;
*
* Note that if data from one knee is missing, the *;
* corresponding by-person variable is calculated based on *;
* the non-missing knee. That is, if the participant had a *;
* total knee replacement (and therefore no x-ray) for his *;
* right knee, but had complete data indicating symptomatic *;
* OA for the left knee, he will end up with: *;
*   Right knee: P01RSXKOA=.P (prosthesis) *;
*   Left knee: P01LSXKOA= 1 (symptomatic OA) *;
*   By person: P01SXKOA = 2 (symp OA in left knee only) *;
*
* Revisions: *;
*   Renamed SXOA (from beta release) to SXKOA *;
*   Added variables LSXKOA, RSXKOA, XRKOA, LXRKOA, RXRKOA, *;
*   RXRKOA2, LXRKOA2, RKSX, LKSX 01/20/06 FH *;
*
*   Corrected summary *;
*   Changed code for P01RKSX and P01LKSX to account for when *;
*   input variables are answered don't know and when answers *;
*   are not expected *;
*   Added clean-up step at the end of program 10/04/06 ES *;
*
*   [Alert 0.2.1] *;
*   Handling of refused changed from 77 to .R and don't know *;
*   from 88 to .D to match latest conventions. *;
*
*   Updated to use global macro for all years for calculation of *;
*   P01LKSX, P01RKSX. The macro code's assignment of output data *;
*   when input variables are answered don't know and when answers *;
*   are not expected is different from the previous baseline *;
*   code. Several participants' values for P01LKSX, P01RKSX have *;
*   changed as a result. *;
*   Some participants have had new reading data entered since *;
*   the last baseline release, so some values may have *;
*   changed. 03/21/07 ES *;

```

```

*****;

data tempfh;
  merge svxrd(in=insvxr)
    daf.sv(keep=id p01kpnr12 p01kpnr p01kpnl12 p01kpnl p01krsr p01krs1
      p01kpnr30 p01kpnl30 p01kpnr30m p01kpnl30m);
  by id;
  if insvxr;

*****;
* Indicators of symptomatic xray knee OA (definite osteophytes *;
* and pain most days of a month in past year) *;
*****;

*By person*;
  if (p01svrkost=2 and (p01kpnr=1 or p01kpnr30m=1)) and
    (p01svlkost=2 and (p01kpnl=1 or p01kpnl30m=1)) then P01SXKOA=3; /*Both*/
  else
    if (p01svrkost=2 and (p01kpnr=1 or p01kpnr30m=1)) then P01SXKOA=1; /*Right only*/
  else
    if (p01svlkost=2 and (p01kpnl=1 or p01kpnl30m=1)) then P01SXKOA=2; /*Left only*/
  else
    P01SXKOA=0; /*Neither*/

*Set to .R:Refused if ppt refused to answer pain questions*;
if p01kpnr=.R or p01kpnl=.R then P01SXKOA=.R; /*Refused*/
else
  /*Set to missing if pain or xray data is missing for BOTH knees*/
  if not(P01SXKOA in (3,1,2,.R)) and (((p01kpnr12=1 and (p01kpnr not in (1,0,.D,.R)) o
r
    (p01kpnr12 not in (1,0,.D,.R))) and ((p01kpnl12=1 and p01kpnl not in (1,0,.D,.R)) or
    (p01kpnl12 not in (1,0,.D,.R)))) or (p01svrkost<0 and p01svlkost<0)
    then P01SXKOA=.M; /*Missing*/

*By knee*;
  if (p01svrkost=2 and (p01kpnr=1 or p01kpnr30m=1)) then P01RSXKOA=1; /*Right knee*/
  else
    P01RSXKOA=0;
  if p01kpnr=.R then P01RSXKOA=.R; /*Refused*/
  else
    if not(P01RSXKOA in (1,.R)) and (p01kpnr12=1 and (p01kpnr not in (1,0,.D,.R))) or
    (p01kpnr12 not in (1,0,.D,.R)) or (p01svrkost<0) then P01RSXKOA=.M; /*Missing*/
  if p01svxrrkr=1 then P01RSXKOA=.P; /*Knee replacement
*/

  if (p01svlkost=2 and (p01kpnl=1 or p01kpnl30m=1)) then P01LSXKOA=1; /*Left knee*/
  else
    P01LSXKOA=0;
  if p01kpnl=.R then P01LSXKOA=.R; /*Refused*/
  else
    if not(P01LSXKOA in (1,.R)) and (p01kpnl12=1 and (p01kpnl not in (1,0,.D,.R))) or
    (p01kpnl12 not in (1,0,.D,.R)) or (p01svlkost<0) then P01LSXKOA=.M; /*Missing*/
  if p01svxrlkr=1 then P01LSXKOA=.P; /*Knee replacement
*/

*****;
* Indicators of xray knee OA (definite osteophytes) *;
*****;

*By person*;
  if (p01svrkost=2 and p01svlkost=2) then P01XRKOA=3; /*Both*/
  else
    if (p01svrkost=2) then P01XRKOA=1; /*Right only*/
  else
    if (p01svlkost=2) then P01XRKOA=2; /*Left only*/
  else
    P01XRKOA=0; /*Neither*/
  if (p01svrkost<0 and p01svlkost<0) then P01XRKOA=.M; /*Both knees missing*/

*By knee*;
  if p01svrkost=2 then P01RXRKO=1; /*Right knee*/

```

```

        else P01RXRKOA=0;
if p01svlkost=2 then P01LXRKOA=1; /*Left knee*/
        else P01LXRKOA=0;

if p01svrkost not in (0,1,2) then P01RXRKOA=.M; /*Missing*/
if p01svlkost not in (0,1,2) then P01LXRKOA=.M;

if p01svxrrkr=1 then P01RXRKOA=.P;          /*Knee replacement*/
if p01svxrlkr=1 then P01LXRKOA=.P;

*****;
* Levels of xray knee OA findings (osteophytes and maximum of medial and lateral JSN) *;
*****;
*Right knee*;
    if p01svrkost=0 and (max(p01svrkjms,p01svrkjls)=0) then P01RXRKOA2=0; /*No osteophytes,
no JSN*/
        else
            if p01svrkost=0 and (max(p01svrkjms,p01svrkjls)>0) then P01RXRKOA2=1; /*No osteophytes,
definite JSN*/
                else
                    if p01svrkost=1 and (max(p01svrkjms,p01svrkjls)=0) then P01RXRKOA2=2; /*Possible osteoph
ytes, no JSN*/
                        else
                            if p01svrkost=1 and (max(p01svrkjms,p01svrkjls)>0) then P01RXRKOA2=3; /*Possible osteoph
ytes, definite JSN*/
                                else
                                    if p01svrkost=2 and (max(p01svrkjms,p01svrkjls)=0) then P01RXRKOA2=4; /*Definite osteoph
ytes, no JSN*/
                                        else
                                            if p01svrkost=2 and (max(p01svrkjms,p01svrkjls)=1) then P01RXRKOA2=5; /*Definite osteoph
ytes, mild-moderate JSN*/
                                                else
                                                    if p01svrkost=2 and (max(p01svrkjms,p01svrkjls)=2) then P01RXRKOA2=6; /*Definite osteoph
ytes, severe JSN*/

if p01svxrrkr=1 then P01RXRKOA2=.P;          /*Knee replacement*/
        else
            if p01svrkost not in (0,1,2) or max(p01svrkjms,p01svrkjls) not in (0,1,2) then P01RX
RKOA2=.M;

*Left knee*;
    if p01svlkost=0 and (max(p01svlkjms,p01svlkjls)=0) then P01LXRKOA2=0; /*No osteophytes,
no JSN*/
        else
            if p01svlkost=0 and (max(p01svlkjms,p01svlkjls)>0) then P01LXRKOA2=1; /*No osteophytes,
definite JSN*/
                else
                    if p01svlkost=1 and (max(p01svlkjms,p01svlkjls)=0) then P01LXRKOA2=2; /*Possible osteoph
ytes, no JSN*/
                        else
                            if p01svlkost=1 and (max(p01svlkjms,p01svlkjls)>0) then P01LXRKOA2=3; /*Possible osteoph
ytes, definite JSN*/
                                else
                                    if p01svlkost=2 and (max(p01svlkjms,p01svlkjls)=0) then P01LXRKOA2=4; /*Definite osteoph
ytes, no JSN*/
                                        else
                                            if p01svlkost=2 and (max(p01svlkjms,p01svlkjls)=1) then P01LXRKOA2=5; /*Definite osteoph
ytes, mild-moderate JSN*/
                                                else
                                                    if p01svlkost=2 and (max(p01svlkjms,p01svlkjls)=2) then P01LXRKOA2=6; /*Definite osteoph
ytes, severe JSN*/

if p01svxrlkr=1 then P01LXRKOA2=.P;          /*Knee replacement*/
        else
            if p01svlkost not in (0,1,2) or max(p01svlkjms,p01svlkjls) not in (0,1,2) then P01LX
RKOA2=.M;

```

```

*****;
* Levels of knee symptom status (none, infrequent, frequent) *;
*****;

%include "&path\macros\RKSX_LKSXG.sas";

run;
data SXOA;
    set tempfh(keep=id P01LSXKOA P01RSXKOA P01SXKOA
                    P01LXRKOA P01RXRKOA P01XRKOA
                    P01LXRKOA2 P01RXRKOA2
                    P01LKSX P01RKSX);
run;

*clean-up;
proc datasets library=work memtype=data;
    delete tempfh;
quit;

```

```

*****;
* Study: OAI *;
* Source name: WKHRCVG.sas *;
* Source type: Program *;
* Author: Maurice Dockrell *;
* Created: 10/03/2006 *;
* Input: &_lib..&Mmo (Clinic visit workbook) *;
* Output: &_v.WKHR7CV: How many hours did you work in past 7 days? *;
* Summary: Calculating number of hours worked in past 7 days from *;
* Enrollment Visit Workbook p19b *;
*
* Follow Up Visit Workbook p35Qb *;
* by assuming 0 hours worked if didn't work in last 7 days, *;
* 1 hour worked if the participant said worked, but less *;
* than 1 hour. Otherwise takes the value of the variable *;
* &_v.WORKHR7 *;
* Revisions: Corrected comments to include baseline MD 04/09/07 *;
*****;

* This code is written for use with data from any dataset, so it uses
  generic macro variables to take the place of variable prefixes that change from year
  to year, as follows:
    &_v. stands for the variable prefixes V00,V01, V02, etc;

*If didnt work in last 7 days, then no hours;
if &_v.WORK7=0 then &_v.WKHR7CV=0;
*If entire question is missing, refused, or dont know, then set special missing values
based on parent question;
  else if &_v.WORK7<=.z then &_v.WKHR7CV=&_v.WORK7;

*If worked, then see if hours filled in on &_v.WORKHR7 or less than 1 hour or dont know comp
leted
  on &_v.WORK1HR;
  else if &_v.WORK7=1 then do;
    if &_v.WORK1HR not in (1,.D) then &_v.WKHR7CV=&_v.WORKHR7;
*Set less than 1 hour to 1 hour;
    else if &_v.WORK1HR=1 then &_v.WKHR7CV=1;
*Otherwise, set missing values based on sub-question missing values;
    else if &_v.WORK1HR<=.z then &_v.WKHR7CV=&_v.WORK1HR;
*Otherwise set missing values based on missing values from &_v.WORKHR7;
    else if &_v.WORKHR7<=.z then &_v.WKHR7CV=&_v.WORKHR7;
  end;

```

```

*****;
* Study: OAI *;
* Source name: WTGA.sas *;
* Source type: Program *;
* Author: Laura Akin *;
* Created: 03/31/2005 *;
* Input: DAF.IEI (Initial Eligibility Interview) *;
*      &_v.AGESR, &_v.WTLB, &_v.SEX *;
* Output:      &_v.WTGA:  No if : Female, 45-69, <=170 *;
*                        Yes if: Female, 45-69,>170 *;
*                        No if : Female, 70-79,<=180 *;
*                        Yes if: Female, 70-79,>180 *;
*                        No if : Male, 45-69, <=205 *;
*                        Yes if: Male, 45-69, >205 *;
*                        No if : Male, 70-79, <=215 *;
*                        Yes if: Male, 70-79, >215 *;
* *;
* Summary:  Calculating Weight group by Gender and Age *;
*            from the IEI variables &_v.AGESR, &_v.WTLB, *;
*            &_v.SEX (IEI page 2-3) and creating *;
*            a Yes/No variable indicating whether or not a *;
*            participant is above the risk factor weight *;
*            cutoff for his or her age/sex *;
* *;
* Revisions: Updated to use macro variables MD 04/02/07 *;
*****;

```

****Some ppts are 44 yrs of age (but will be 45 by EV) and are included in the 45-69 age group;

****Setting to missing where any component value is missing:

```

if any component has .D then set to .D
  else if any component has .R then set to .R
  else if any component has .A then set to .A
  else if any component has .N then set to .N
  else if any component is missing set to .M

```

****;

```

if &_v.AGESR=.D or &_v.WTLB=.D or &_v.SEX=.D then &_v.WTGA=.D;
else if &_v.AGESR=.R or &_v.WTLB=.R or &_v.SEX=.R then &_v.WTGA=.R;
else if &_v.AGESR=.A or &_v.WTLB=.A or &_v.SEX=.A then &_v.WTGA=.A;
else if &_v.AGESR=.N or &_v.WTLB=.N or &_v.SEX=.N then &_v.WTGA=.N;
else if &_v.AGESR<=.z or &_v.WTLB<=.z or &_v.SEX<=.z then &_v.WTGA=.M;

```

****Calculating Yes/No based on age, weight, gender: Yes if above weight cutoff, otherwise No;

```

else if ((44 <= &_v.AGESR <= 69) and (&_v.WTLB<=170) and (&_v.SEX=2)) then &_v.WTGA=0
;
else if ((44 <= &_v.AGESR <= 69) and (&_v.WTLB > 170) and (&_v.SEX=2)) then &_v.WTGA=
1;
else if ((70 <= &_v.AGESR <= 79) and (&_v.WTLB <= 180) and (&_v.SEX=2)) then &_v.WTGA=
0;
else if ((70 <= &_v.AGESR <= 79) and (&_v.WTLB > 180) and (&_v.SEX=2)) then &_v.WTGA=1
;
else if ((44 <= &_v.AGESR <= 69) and (&_v.WTLB<= 205) and (&_v.SEX=1)) then &_v.WTGA=0
;
else if ((44 <= &_v.AGESR <= 69) and (&_v.WTLB > 205) and (&_v.SEX=1)) then &_v.WTGA=1
;
else if ((70 <= &_v.AGESR <= 79) and (&_v.WTLB <= 215) and (&_v.SEX=1)) then &_v.WTGA=
0;
else if ((70 <= &_v.AGESR <= 79) and (&_v.WTLB > 215) and (&_v.SEX=1)) then &_v.WTGA=1
;

```